Introduction to Python

Why learn Python?

Python is increasingly the most popular choice of programming language for data analysts because it is designed to be simple, efficient, and easy to read and write.

There are many open source software and libraries that use Python and data analysis tools built on them.

We will use Python to learn programming and explore fundamental programming concepts of commands, variables, decisions, repetition, and events.

What is Python?

Python is a general, high-level programming language designed for code readability and simplicity.

Python is available for free as open source and has a large community supporting its development and associated tools.

Python was developed by Guido van Rossum and first released in 1991. Python 2.0 was released in 2000 (latest version 2.7), and a backwards-incompatible release Python 3 was in 2008.

- Our coding style will be Python 3 but most code will also work for Python 2.

Python Language Characteristics

Python supports:

- dynamic typing – types can change at run-time
- multi-paradigm – supports procedural, object-oriented, functional styles
- auto-memory management and garbage collection
- extendable – small core language that is easily extendable

Python core philosophies (by Tim Peters: https://www.python.org/dev/peps/pep-0020/)

- Beautiful is better than ugly
- Explicit is better than implicit
- Simple is better than complex
- Complex is better than complicated
- Readability counts

Some Quotes

"If you can't write it down in English, you can't code it."
-- Peter Halpern

"If you lie to the computer, it will get you."
-- Peter Farrar

Introduction to Programming

An ***algorithm*** is a precise sequence of steps to produce a result. A ***program*** is an encoding of an algorithm in a ***language*** to solve a particular problem.

There are numerous languages that programmers can use to specify instructions. Each language has its different features, benefits, and usefulness.

The goal is to understand fundamental programming concepts that apply to all languages.

Python: Basic Rules

To program in Python you must follow a set of rules for specifying your commands. This set of rules is called a ***syntax***.

- Just like any other language, there are rules that you must follow if you are to communicate correctly and precisely.

Important general rules of Python syntax:

- Python is ***case-sensitive***.
- Python is particular on whitespace and indentation.
- The end of command is the end of line. There is no terminator like a semi-colon.
- Use four spaces for indentation whenever in a block.

```
def spam():
    eggs = 12
    return eggs
print spam()
```

Comments

Comments are used by the programmer to document and explain the code. Comments are ignored by the computer. Two types:

- 1) One line comment: put "#" before the comment and any characters to the end of line are ignored by the computer.
- 2) Multiple line comment: put """ at the start of the comment and """ at the end of the comment. The computer ignores everything between the start and end comment indicators.

Example:

```
# Single line comment
print (1)  # Comment at end of line
""" This is a
multiple line
comment """
```

Python Programming

A Python program, like a book, is read left to right and top to bottom.

Each command is on its own line.

```
# Sample Python program
name = "Joe"
print("Hello")
print("Name: "+name)
```

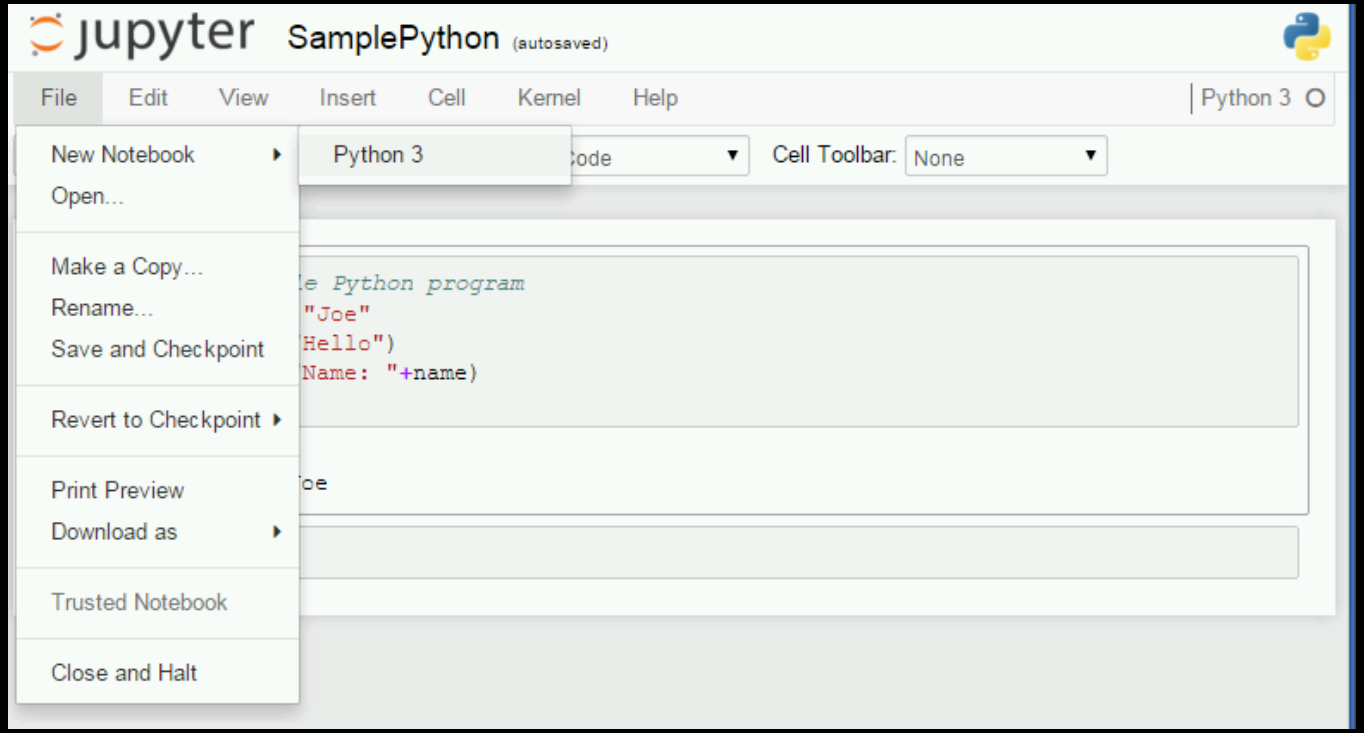
Flow of Execution
Start at first statement at top and proceed down executing each statement

A user types in a Python program in a text editor or development environment and then runs the program.

Python Editor - jupyter

jupyter is a graphical, browser-based editor for Python.

To create a new notebook, select `File`, `New Notebook`, `Python3`.



Python Editor – jupyter notebook

Button to run code (shortcut is Ctrl+Enter)

jupyter SamplePython (autosaved)

File Edit View Insert Cell Kernel Help Python 3

Code Cell Toolbar: None

In [2]:
```
# Sample Python program
name = "Joe"
print("Hello")
print("Name: "+name)
```

Python code

```
Hello
Name: Joe
```

Program output

In [ ]:

Python: Hello World!

Simplest program:

```
print("Hello World!")
```

The `print` function will print to the terminal (standard output) whatever data (number, string, variable) it is given.

Try it: Python Printing

Question 1: Write a Python program that prints "I am fantastic!".

Question 2: Write a Python program that prints these three lines:

```
I know that I can program in Python.
I am programming right now.
My awesome program has three lines!
```

Python Question

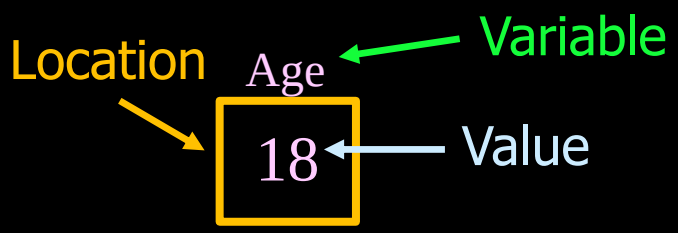
Question: How many of the following statements are **TRUE**?

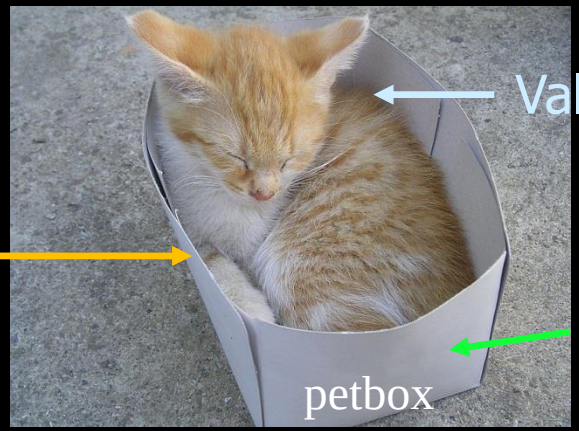
1) Python is case-sensitive.
2) A command in Python is terminated by a semi-colon.
3) Indentation does not matter in Python.
4) A single line comment starts with `"""`.
5) The `print` command prints to standard input.

A) 0 **B)** 1 **C)** 2 **D)** 3 **E)** 4

Variables

A ***variable*** is a name that refers to a location that stores a data value.





IMPORTANT: The *value* at a location can change using initialization or assignment.

Variable Assignment

Assignment using an **=** sets the value of a variable.

Example:

```
num = 10
message = "Hello world!"
```

num

10

message

Hello world!

Python Variables

To create a variable in Python, you must only provide a name.

- A variable type is dynamic. It can store numbers, strings, or Boolean at any time.

Example:

```
val = 5
val = "Hello"
```

Boolean values can be either `True` or `False`. Note case matters.

```
isAwesome = True
isAwesome = False
```

Variable Rules

Variables are a name that ***must begin with a letter*** and cannot contain spaces.

Variables are created when they are first used. There is no special syntax to declare (create) a variable.

Variable names **ARE** case-sensitive. Numbers are allowed (but not at the start). Only other symbol allowed is underscore ('_');

A programmer picks the names for variables, but try to make the names meaningful and explain their purpose.

Avoid naming variables as reserved words. A ***reserved word*** has special meaning in the language.

- e.g. `if`, `for`, `else`

Python Variables Question

Question: How many of the following variable names are valid?

1) `name`
2) `string2`
3) `2cool`
4) `under_score`
5) `spaces name`
6) `else`

A) 0 **B)** 1 **C)** 2 **D)** 3 **E)** 4

Python Math Expressions

Math ***expressions*** in Python:

Operation	Syntax	Example
Add	+	5 + 3
Subtract	-	10 – 2
Multiply	*	5 * 3
Divide	/	9 / 4
Modulus	%	9 % 4 (answer is 1)
Exponent	**	5 ** 2 (answer is 25)

Expressions - Operator Precedence

Each operator has its own priority similar to their priority in regular math expressions:

- 1) Any expression in parentheses is evaluated first starting with the inner most nesting of parentheses.
- 2) Exponents
- 3) Multiplication and division (*, /, %)
- 4) Addition and subtraction (+,-)

Python Expressions Question

Question: What is the value of this expression:

```
8 ** 2 + 12 / 4 * (3 - 1) % 5
```

A) 69 **B)** 65 **C)** 36 **D)** 16 **E)** 0

Try it: Python Variables and Expressions

Question 1: Write a program that prints the result of `35 + 5 * 10`.

Question 2: Write a program that uses at least 3 operators to end up with the value `99`.

Question 3: Write a program that has a variable called `name` with the value of your name and a variable called `age` storing your age. Print out your name and age using these variables.

Strings

Strings are sequences of characters that are surrounded by either single or double quotes.

- Use \ to escape ' E.g. There\'s
- Can use triple double quotes """ for a string that spans multiple lines.

Example:

```
name = "Joe Jones"
storeName = 'Joe\'s Store'
print("""String that is really long
with multiple lines
   and spaces is perfectly fine""")
```

Python String Indexing

Individual characters of a string can be accessed using square brackets (`[]`) with the first character at index 0.

Example:

```
str = "Hello"
print(str[1])        # e
print("ABCD"[0])     # A
print(str[-1])       # o
# Negative values start at end and go backward
```

Rules for Strings in Python

Must be surrounded by single or double quotes.

Can contain most characters except enter, backspace, tab, and backslash.

- These special characters must be escaped by using an initial "\".
- e.g. `\n` – new line, `\'` – single quote, `\\` - backslash, `\"` – double quote
- A string in raw mode (`r` before quote) will ignore backslash escape. May be useful if data contains escapes. Example: `st = r"slash\there\"`

Double quoted strings can contain single quoted strings and vice versa.

Any number of characters is allowed.

The minimum number of characters is zero `""`, which is called the *empty string*.

String *literals* (values) have the quotation marks removed when displayed.

Python Strings Question

Question: How many of the following are valid Python strings?

1) `""`
2) `''`
3) `"a"`
4) `" "`
5) `"""`
6) `"Joe\' Smith\""`

A) 1 **B)** 2 **C)** 3 **D)** 4 **E)** 5

Python String Functions

```
st  = "Hello"
st2 = "Goodbye"
```

Operation	Syntax	Example	Output
Length	`len()`	`len(st)`	`5`
Upper case	`upper()`	`st.upper()`	`HELLO`
Lower case	`lower()`	`st.lower()`	`hello`
Convert to a string	`str()`	`str(9)`	`"9"`
Concatenation	`+`	`st1 + st2`	`HelloGoodbye`
Substring	`[]`	`st[0:3]` `st[1:]`	`Hel` `ello`
String to int	`int()`	`int("99")`	`99`

String Operators: Concatenation

The ***concatenation operator*** is used to combine two strings into a single string. The notation is a plus sign '**+**'.

Example:

```
st1 = "Hello"
st2 = "World!"
st3 = st1 + st2 # HelloWorld!
print(st1+st1)
num = 5
print(st1+str(num)) # Hello5
# Must convert number to string before
# concatenation
```

String Concatenation Question

Question: What is the output of this code?

```
st1 = "Hello"
st2 = "World!"
num = 5
print(st1 + str(num) + " " + st2)
```

A) `Error`

B) `Hello5World!`

C) `Hello5 World!`

D) `Hello 5 World!`

Substring

The ***substring*** function will return a range of characters from a string.

Syntax:

```
st[start:end]   # start is included, end is not
                # first character is index 0
```

Examples:

```
st = "Fantastic"
print(st[1])          # a
print(st[0:6])        # Fantas
print(st[4:])         # astic
print(st[:5])         # Fanta
print(st[-6:-2])      # tast
```

Substring Question

Question: What is the output of this code?

```
st = "ABCDEFG"
print(st[1] + st[2:4] + st[3:] + st[:4])
```

A) ABCDCDEFGABCD

B) ABCDEFGABC

C) ACDDEFGABCD

D) BCDDEFGABCD

E) BCDECDEFGABC

Split

The ***split*** function will divide a string based on a separator.

Examples:

```
st = "Awesome coding! Very good!"
print(st.split())
# ['Awesome', 'coding!', 'Very', 'good!']
print(st.split("!"))
# ['Awesome coding', ' Very good', '']
st = 'data,csv,100,50,,25,"use split",99'
print(st.split(","))
# ['data', 'csv', '100', '50', '', '25',
#   '"use split"', '99']
```

Try it: Python String Variables and Functions

Question 1: Write a Python program that prints out your name and age stored in variables like this:

```
Name: Joe
Age: 25
```

Question 2: Write a Python program that prints out the first name and last name of `Steve Smith` like below. You must use substring.

- Bonus challenge: Use `find()` function so that it would work with any name.

```
First Name: Steve
Last Name: Smith
```

Print Formatting

The `print` method can accept parameters for formatting.

```
print("Hi", "Amy", ", your age is", 21)
print("Hi {}, your age is {}".format("Amy",21))
```

This is one of the most obvious changes between Python 2:

```
print "Hello"
```

and Python 3:

```
print("Hello")
```

Python Date and Time

Python supports date and time data types and functions.

First, import the `datetime` module:

```
from datetime import datetime
```

Functions:

```
now = datetime.now()
print(now)
current_year = now.year
current_month = now.month
current_day = now.day
print("{}-{}-{} {}:{}:{}".format(now.year, now.month,
now.day, now.hour, now.minute, now.second))
```

Python Clock

Python `time()` function returns the current time in seconds:

```
import time
startTime = time.time()
print("Start time:", startTime)
print("How long will this take?")
endTime = time.time()
print("End time:", endTime)
print("Time elapsed:", endTime-startTime)
```

Python Input

To read from the keyboard (standard input), use the method `input`:

```
name = input("What's your name?")
print(name)
age = input("What's your age?")
print(age)
```

Prompt for value from user

print out value received

- Note in Python 2 the method is called `raw_input()`.

Try it: Python Input, Output, and Dates

Question 1: Write a program that reads a name and prints out the name, the length of the name, the first five characters of the name.

Question 2: Print out the current date in `YYYY/MM/DD` format.

Comparisons

A ***comparison operator*** compares two values. Examples:

- `5 < 10`
- `N > 5  # N is a variable. Answer depends on what is N.`

Comparison operators in Python:

- `>` - Greater than
- `>=` - Greater than or equal
- `<` - Less than
- `<=` - Less than or equal
- `==` - Equal (Note: Not "=" which is used for assignment!)
- `!=` - Not equal

The result of a comparison is a ***Boolean value*** which is either **`True`** or **`False`**.

Conditions with `and`, `or`, `not`

A ***condition*** is an expression that is either `True` or `False` and may contain one or more comparisons. Conditions may be combined using: `and`, `or`, `not`.

- order of evaluation: `not`, `and`, `or` May change order with parentheses.

Operation	Syntax	Examples	Output

AND
(True if both are True)
	`and`	`True and True`
		`False and True`
		`False and False`
			`True`
			`False`
			`False`

OR
(True if either or both are True)
	`or`	`True or True`
		`False or True`
		`False or False`
			`True`
			`True`
			`False`

NOT **(Reverses: e.g. True becomes False)**	`not`	`not True` `not False`	`False` `True`

Condition Examples

```
n = 5
v = 8
print(n > 5)                          # False
print(n == v)                         # False
print(n != v)                         # True
print(n == v and n+4>v)               # False
print(n == v or n+4>v)                # True
print(n+1 == v-2 or not v>4)          # True
```

Python Condition Question

Question: How many of the following conditions are **TRUE**?

1) `True and False`
2) `not True or not False`
3) `3 > 5 or 5 > 3 and 4 != 4`
4) `(1 < 2 or 3 > 5) and (2 == 2 and 4 != 5)`
5) `not (True or False) or True and (not False)`

A) 0 **B)** 1 **C)** 2 **D)** 3 **E)** 4

Decisions

Decisions allow the program to perform different actions based on conditions. Python decision syntax:

```
if condition:
    statement
```

Done if condition is `True`

```
if condition:
    statement
else:
    statement
```

Done if condition is `False`

- The statement after the `if` condition is only performed if the condition is `True`.
- If there is an `else`, the statement after the else is done if condition is `False`.
- Indentation is important! Remember the colon!

Decisions `if/elif` Syntax

If there are more than two choices, use the `if/elif/else` syntax:

```
if condition:
    statement
elif condition:
    statement
elif condition:
    statement
else:
    statement
```

```
if n == 1:
    print("one")
elif n == 2:
    print("two")
elif n == 3:
    print("three")
else:
    print("Too big!")
print("Done!")
```

Decisions: Block Syntax

Statements executed after a decision in an `if` statement are indented for readability. This indentation is also how Python knows which statements are part of the block of statements to be executed.

- If you have more than one statement, make sure to indent them. Be consistent with either using tabs or spaces. Do not mix them!

```
if age > 19 and name > "N":
    print("Not a teenager")
    print("Name larger than N")
else:
    print("This is statement #1")
    print(" and here is statement #2!")
```

Question: Decisions

Question: What is the output of the following code?

```
n = 3
if n < 1:
    print("one")
elif n > 2:
    print("two")
elif n == 3:
    print("three")
```

A) nothing **B)** `one` **C)** `two` **D)** `three`

Question: Decisions (2)

Question: What is the output of the following code?

```
n = 3
if n < 1:
   print("one")
elif n > 2
   print("two")
else:
   print("three")
```

A) nothing

B) one

C) two

D) three

E) error

Question: Decisions (3)

Question: What is the output of the following code?

```
n = 1
if n < 1:
    print("one")
elif n > 2:
    print("two")
else:
    print("three")
print("four")
```

A) nothing

B) one
four

C) three

D) three
four

E) error

Question: Decisions (4)

Question: What is the output of the following code?

```
n = 0
if n < 1:
    print("one")
    print("five")
elif n == 0:
    print("zero")
else:
    print("three")
print("four")
```

A) nothing

B)
```
one
four
```

C)
```
one
five
four
```

D)
```
one
five
zero
four
```

E) error

Try it: Decisions

Question 1: Write a Python program that asks the user for a number then prints out if it is even or odd.

Question 2: Write a Python program that asks the user for a number between 1 and 5 and prints out the word for that number (e.g. 1 is one). If the number is not in that range, print out error.

Loops and Iteration

A ***loop*** repeats a set of statements multiple times until some condition is satisfied.

- Each time a loop is executed is called an ***iteration***.

A `for` loop repeats statements a number of times.

A `while` loop repeats statements while a condition is `True`.

The `while` Loop

The most basic looping structure is the ***while*** loop.

A while loop continually executes a set of statements **while** a condition is true.

Syntax:

```
while condition:
    statements
```

Example:

```
n = 1
while n <= 5:
    print(n)
    n = n + 1    # Shorthand: n += 1
```

Question: What does this print?

Question: `while` Loop

Question: What is the output of the following code?

```
n = 4
while n >= 0:
    n = n - 1
    print(n)
```

A) numbers 3 to -1 **B)** numbers 3 to 0 **C)** numbers 4 to 0

D) numbers 4 to -1 **E)** numbers 4 to infinity

Question: `while` Loop (2)

Question: What is the output of the following code?

```
n = 1
while n <= 5:
    print(n)
n = n + 1
```

A) nothing **B)** numbers 1 to 5 **C)** numbers 1 to 6 **D)** lots of 1*s*

The `for` Loop

A `for` loop repeats statements a given number of times.

Python `for` loop syntax:

```
for i in range(1,6):
    print(i)
```

Up to but not including ending number

Starting number

Using `range`

The basic form of range is:

```
range(start,end)
```

- start is inclusive, end is not inclusive
- default increment is 1

May also specify an increment:

```
range(start, end, increment)
```

or just the end:

```
range(end)
```

For Loop and While Loop

The `for` loop is like a short-hand for the `while` loop:

```
i=0
while i < 10:
    print(i)
    i += 1
```

```
for i in range(0, 10, 1):
    print(i)
```

Common Problems – Infinite Loops

Infinite loops are caused by an incorrect loop condition or not updating values within the loop so that the loop condition will eventually be false.

Example:

```
n = 1
while n <= 5:
    print(n)
    # Forgot to increase n -> infinite loop
```

Common Problems – Off-by-one Error

The most common error is to be "***off-by-one***". This occurs when you stop the loop one iteration too early or too late.

Example:

- This loop was supposed to print 0 to 10, but it does not.

```
for i in range(0,10):
    print(i)
```

Question: How can we fix this code to print 0 to 10?

Question: for Loop

Question: How many numbers are printed with this loop?

```
for i in range(1,10):
    print(i)
```

A) 0 **B)** 9 **C)** 10 **D)** 11 **E)** error

Question: for Loop

Question: How many numbers are printed with this loop?

```
for i in range(11,0):
    print(i)
```

A) 0 **B)** 9 **C)** 10 **D)** 11 **E)** error

Try it: `for` Loops

Question 1: Write a program that prints the numbers from 1 to 10 then 10 to 1.

Question 2: Write a program that prints the numbers from 1 to 100 that are divisible by 3 and 5.

Question 3: Write a program that asks the user for 5 numbers and prints the maximum, sum, and average of the numbers.

Lists Overview

A ***list*** is a collection of data items that are referenced by index.

- Lists in Python are similar to arrays in other programming languages

A list allows multiple data items to be referenced by one name and retrieved by index.

Python list:

```
data = [100, 200, 300, 'one', 'two', 600]
```

0	1	2	3	4	5

list variable name

Indexes

Retrieving Items from a List

Items are retrieved by index (starting from 0) using square brackets:

```
data = [100, 200, 300, 'one', 'two', 600]
print(data[0])              # 100
print(data[4])              # 'two'
print(data[6])              # error – out of range
print(data[len(data)-1])    # 600
print(data[-1])             # 600
print(data[2:4])            # [300, 'one']

# Create an empty list:
emptyList = []
```

List Operations

```
data = [1, 2, 3, 5]
lst = []
```

Operation	Syntax	Examples	Output
Add item	*list*.append(*val*)	data.append(1)	[1, 2, 3, 5, 1]
Insert item	*list*.insert(*idx*,*val*)	data.insert(3,4)	[1, 2, 3, 4, 5]
Remove item	*list*.remove(*val*)	data.remove(5)	[1, 2, 3]
Update item	*list*[*idx*]=*val*	lst[0]=10	[10]
Length of list	len(*list*)	len(data)	4
Slice of list	*list*[x:y]	data[0:3]	[1, 2, 3]
Find index	*list*.index(*val*)	data.index(5)	3
Sort list	*list*.sort()	data.sort()	[1, 2, 3, 5]

List Details

If you provide an index outside of the valid range, Python will return an index error.

To sort in reverse order, do this:

```
data.sort(reverse=True)
```

For loops are used to iterate though items in a list:

```
for v in data:
   print(v)
```

Advanced: Python Lists Comprehensions

List comprehensions build a list using values that satisfy a criteria.

```
evenNums100 = [n for n in range(101) if n%2==0]
```

Equivalent to:

```
evenNums100 = []
for n in range(101):
   if n%2==0:
       evenNums100.append(n)
```

Advanced: Python Lists Slicing

List slicing allows for using range notation to retrieve only certain elements in the list by index. Syntax:

```
list[start:end:stride]
```

Example:

```
data = list(range(1,11))
print(data)     # [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
print(data[1:8:2])  # [2, 4, 6, 8]
print(data[1::3])   # [2, 5, 8]
```

Question: List

Question: At what index is item with value 3?

```
data = [1, 2, 3, 4, 5]
data.remove(3)
data.insert(1, 3)
data.append(2)
data.sort()
data = data[1:4]
```

A) 0 **B)** 1 **C)** 2 **D)** 3 **E)** not there

Try it: Lists

Question 1: Write a program that puts the numbers from 1 to 10 in a list then prints them by traversing the list.

Question 2: Write a program that will multiply all elements in a list by 2.

Question 3: Write a program that reads in a sentence from the user and splits the sentence into words using split(). Print only the words that are more than 3 characters long. At the end print the total number of words.

Python Dictionary

A ***dictionary*** is a collection of key-value pairs that are manipulated using the key.

```
dict = {1:'one', 2:'two', 3:'three'}
print(dict[1])          # one
print(dict['one'])      # error - key not found
if 2 in dict:           # Use in to test for key
   print(dict[2])       # two
dict[4] = 'four'        # Add 4:'four'
del dict[1]             # Remove key 1
dict.keys()             # Returns keys
dict.values()           # Returns values
```

Question: Dictionary

Question: What is the value printed?

```
data = {'one':1, 'two':2, 'three':3}
data['four'] = 4
sum = 0
for k in data.keys():
   if len(k) > 3:
      sum = sum + data[k]
print(sum)
```

A) 7 **B)** 0 **C)** 10 **D)** 6 **E)** error

Try it: Dictionary

Question: Write a program that will use a dictionary to record the frequency of each letter in a sentence. Read a sentence from the user then print out the number of each letter.

Code to create the dictionary of letters:

```
import string
counts = {}
for letter in string.ascii_uppercase:
    counts[letter] = 0
print(counts)
```

Functions and Procedures

A ***procedure*** (or ***method***) is a sequence of program statements that have a specific task that they perform.

- The statements in the procedure are mostly independent of other statements in the program.

A ***function*** is a procedure that returns a value after it is executed.

We use functions so that we do not have to type the same code over and over. We can also use functions that are built-in to the language or written by others.

★Defining and Calling Functions and Procedures

Creating a function involves writing the statements and providing a ***function declaration*** with:

- a name (follows the same rules as identifiers)
- list of the inputs (called parameters)
- the output (return value) if any

Calling (or executing) a function involves:

- providing the name of the function
- providing the values for all arguments (inputs) if any
- providing space (variable name) to store the output (if any)

Defining and Calling a Function

Consider a function that returns a number doubled:

def Keyword

Function Name

Parameter Name

```
def doubleNum(num):
    num = num * 2
    print("Num: "+num)
    return num
```

Call function by name

Argument

```
n = doubleNum(5)                    # 10
print(str(doubleNum(n)))            # ??
```

Python Built-in Math Functions

```
# Math
import math
print(math.sqrt(25))

# Import only a function
from math import sqrt
print(sqrt(25))

# Print all math functions
print(dir(math))
```

Other Python Built-in Functions

`max`, `min`, `abs`:

```
print(max(3, 5, 2))     # 5
print(min(3, 5, 2))     # 2
print(abs(-4))          # 4
```

`type()` returns the argument data type:

```
print(type(42))         # <class 'int'>
print(type(4.2))        # <class 'float'>
print(type('spam'))     # <class 'str'>
```

Python Random Numbers

Use random numbers to make the program have different behavior when it runs.

```
from random import randint
coin = randint(0, 1)          # 0 or 1
die = randint(1, 6)           # 1 to 6
print(coin)
print(die)
```

Advanced: Python Functions

Python supports functional programming allowing functions to be passed like variables to other functions.

- Lambda functions are functions that do not have a name.

Example:

```
def doFunc(func, val):
   return func(val)

print(doFunc(doubleNum, 10))        # 20
print(doFunc(lambda x: x * 3, 5))   # 15
```

Question: Functions

Question: What is the value printed?

```
def triple(num):
    return num * 3

n = 5
print(triple(n)+triple(2))
```

A) 0 **B)** 6 **C)** 15 **D)** 21 **E)** error

Practice Questions: Functions

1) Write a function that returns the largest of two numbers.

2) Write a function that prints the numbers from 1 to N where N is its input parameter.

Call your functions several times to test that they work.

Conclusion

Python is a general, high-level programming language designed for code readability and simplicity.

Programming concepts covered:

- variables, assignment, expressions, strings, string functions
- making decisions with conditions and `if`/`elif`/`else`
- repeating statements (loops) using `for` and `while` loops
- reading input with `input()` and printing with `print()`
- data structures including lists and dictionaries
- creating and calling functions, using built-in functions (math, random)

Python is a powerful tool for data analysis and automation.

Objectives

- Explain what is Python and note the difference between Python 2 and 3
- Define: algorithm, program, language, programming
- Follow Python basic syntax rules including indentation
- Define and use variables and assignment
- Apply Python variable naming rules
- Perform math expressions and understand operator precedence
- Use strings, character indexing, string functions
- String functions: split, substr, concatenation
- Use Python datetime and clock functions
- Read input from standard input (keyboard)

Objectives (2)

- Create comparisons and use them for decisions with `if`
- Combine conditions with `and`, `or`, `not`
- Use `if`/`elif`/`else` syntax
- Looping with `for` and `while`
- Create and use lists and list functions
- Advanced: list comprehensions, list slicing
- Create and use dictionaries
- Create and use Python functions
- Use built-in functions in math library
- Create random numbers
- Advanced: passing functions, lambda functions